Royal Devon University Healthcare NHS Foundation Trust

Research and Development

Standard Operating Procedure

S42 – Management of Research Samples	
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Controlled document

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It is the responsibility of all users of this SOP to ensure that the correct version is being used. If you are reading this in a paper format please go on-line to confirm you have the latest version.

DISCLAIMER

This generic R&D Standard Operating Procedure (SOP) must be followed unless:

- a study specific SOP exists
- a Departmental SOP dictates a different working practice

Once printed this is an uncontrolled document

Full History			
Version	**Date**	**Reviewed By**	**Reason**
1.1	20/03/2014	-	Minor formatting changes to table on front page
	27/01/2021		S42 V1.1 archived.
2	09/05/2024	Research & Development Manager	Reinstated SOP to combine Work Instruction and SOP

Associated Trust Policies/ Procedural documents:	Waste Management Policy (Eastern Services) Waste Management Policy (Northern Services) Control of Substances Hazardous to Health (COSHH) Policy (Eastern Services) Control of Substances Hazardous to Health Policy (Northern Services) Infection Prevention & Control Policy
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KEY POINTS OF THIS PROCEDURAL DOCUMENT

- Samples should be received, processed, stored and shipped in accordance with the study Protocol and Laboratory Manual, Good Clinical practice (GCP) and local policies.
- Guidance for the handling of samples must also be compatible with UK Health and Safety Executive Regulations.
- The study Principal Investigator (PI) must ensure that samples are processed, stored and shipped according to the study Protocol and/ or study laboratory manual.
- R&D staff must ensure they have undertaken appropriate training and have documented evidence of completed training.
- Personal protective equipment (PPE) such as gloves, aprons and eye protection (safety goggles) must be used at all times when processing body fluids.
- Sample tubes must be correctly labelled according to the sample processing instructions in the study Protocol and/or study laboratory manual.
- Samples must be stored according to the instructions in the study Protocol and/or laboratory manual.

1. INTRODUCTION

As part of a research study, there may be a requirement to collect clinical samples. It is the responsibility of the designated staff member(s) to ensure that these samples are received, processed, stored and shipped in accordance with the study Protocol and Laboratory Manual, Good Clinical practice (GCP) and local policies.

Samples for clinical research may include but are not limited to peripheral blood, sputum, nasal washes, mouth swabs, tissue, stool and urine. The integrity of such samples may be affected by the conditions in which they are collected, transported and stored. Along with this standard operating procedure (SOP), staff must also refer to the study Protocol for more detailed guidance on the conditions required for the storage and transportation of samples for specific research studies. Guidance for the handling of samples must also be compatible with UK Health and Safety Executive Regulations.

2. PURPOSE

The purpose of this (SOP) is to describe the processes for the safe processing, storage and shipment of clinical samples maintained for the purposes of clinical research.

3. SCOPE

This SOP should be used by R&D staff involved in receiving, processing, storing and/or shipping clinical samples used for the purposes of clinical research.

4. DEFINITIONS & ABBREVIATIONS

GCP	Good Clinical Practice
GOG	R&D Governance Oversight Group
PI	Principal Investigator
PPE	Personal protective equipment
R&D	Research & Development
SOP	Standard Operating Procedure
Sponsor	An individual, company, institution or organisation which takes responsibility for the initiation, management and financing of a clinical trial. Sponsorship activities may be delegated to the Investigator, CTU and/ or other organisations as appropriate

5. DUTIES AND RESPONSIBILITIES OF STAFF

It is the responsibility of each study's **Principal Investigator** (PI) to ensure that samples are processed, stored and shipped according to the study Protocol, study laboratory manual and/or local practices/policies. This responsibility may be delegated to an appropriately trained member of staff as documented in the delegation log.

R&D staff involved in receiving, processing, storing and/or shipping research samples must ensure they have undertaken appropriate training and provide documented evidence of this training on request.

6. PROCEDURES

6.1 Processing of samples

6.1.1 Personal protective equipment (PPE) such as gloves, aprons and eye protection (safety goggles) must be used at all times when processing body fluids. Gloves should be removed to complete other tasks in the sample processing areas (e.g. using the telephone) and protective clothing should be removed before exiting the sample processing areas.

6.1.2 Study-specific sample processing instructions should be followed as detailed in the study Protocol and/or study laboratory manual.

6.1.3 Samples must be collected in containers supplied by the Sponsor or as defined by the study Protocol / Laboratory manual if supplied locally.

6.1.4 Plasma, serum or blood clot samples are obtained from whole blood after spinning in a centrifuge and are stored in cryovials supplied by the study Sponsor. Urine samples must be transferred from the collection pot into a designated sample tube using a pipette. Sputum and whole blood samples may be frozen in the blood tube they are collected in. DNA samples collected on mouth swab sticks will be supplied with a sterile bag or pot for the swab to be placed in after collection.

6.1.5 Sample tubes must be correctly labelled according to the sample processing instructions in the study Protocol and/or study laboratory manual. Label details should be cross-referenced with the study subject and study-specific documentation to minimise errors. Labelling must be done whilst still with the participant.

6.1.6 All waste material should be placed in a sharps bin or clinical waste bin as appropriate and in compliance with the waste management policies:

Waste Management Policy (Eastern Services)

Waste Management Policy (Northern Services)

6.1.7 In the event of any spillages, staff should refer to the:

Control of Substances Hazardous to Health (COSHH) Policy and Spillage Procedure (Eastern Services).

Control of Substances Hazardous to Health Policy (Northern Services)

6.2 Storage of samples

6.2.1 Samples must be stored according to the instructions in the study Protocol and/or laboratory manual.

6.2.2 If no study-specific instructions have been provided, samples should be stored in an appropriate storage box suitable for the storage temperature and size of the tubes.

6.2.3 Serum or plasma samples may be stored in sample grid boxes with lids. These boxes must be clearly labelled in waterproof ink with the study name, study number, box number and contact telephone number for the local study team. Other samples must be placed within clear plastic ‘marsupial’ bags and sealed.

6.2.4 Study sample logs should be completed where required to document the processing and storage of samples.

6.2.5 A CRF Freezer Storage Application form must be completed for any samples that will be stored within CRF freezer storage.

6.3 Transporting samples at the Local Site

6.3.1 Samples should be wrapped within a clear plastic bag and sealed. Clear 'marsupial' bags are often used in the clinical environment as these have an outer pocket for any accompanying forms. However, these may not be necessary for some clinical trials. Some samples can be securely transported in an appropriate transport container.

6.3.2 For transport from the point of collection to the centrifuge laboratory or freezer storage, place the wrapped and bagged sample within a leak-proof plastic container and close the lid. Ideally, these boxes must be clear enough to see the contents inside.

6.3.3 In the event of damage, the sample tube must be left inside the clear plastic bag or container unopened and discarded as a whole in a clinical waste bin. Loss of the sample must be recorded in study documentation.

6.4 Shipping samples

6.4.1 Study-specific instructions should be followed as detailed in the study Protocol and/or study laboratory manual for shipping samples to the Sponsor.

7. DISSEMINATION AND TRAINING

7.1 This SOP and associated templates and forms will be uploaded to the Royal Devon website shortly after having been released.

7.2 All staff whose activities are subject to this SOP must ensure that they take time to read and understand the content of this SOP.

7.3 If applicable, a training log within the Investigator Site File/Trial Master File must be completed to document that members of staff have read and understood the contents of this SOP.

8. MONITORING COMPLIANCE AND EFFECTIVENESS OF THIS SOP

8.1 In order to monitor compliance with this SOP, the auditable standards will be monitored as follows:

No.	Minimum Requirements	Evidenced by
1.	Staff involved in receiving, processing, storing and/or shipping research samples have completed and evidence completion of appropriate training.	Check staff training records
2.	Study sample logs have been completed	Check study sample logs in study folder

8.2 Outcomes from audit will be presented to the R&D Governance Oversight Group (GOG) which will monitor any resulting action plans until all issues have been addressed to satisfaction

8.3 Issues identified via the audit process which require escalation will be referred to GOG.

9. ARCHIVING ARRANGEMENTS

9.1 The original of this document will remain with the R&D Quality Assurance Coordinator. An electronic copy will be maintained on the R&D section of the Q-Pulse document management system and a pdf copy on the Royal Devon website.

9.2 Archive copies must be maintained for any documents which have been superseded. Archive copies in electronic format should be retained indefinitely.

10. REFERENCES

Waste Management Policy (Eastern Services)

Waste Management Policy (Northern Services)

Control of Substances Hazardous to Health (COSHH) Policy (Eastern Services)

Control of Substances Hazardous to Health Policy (Northern Services)

Infection Prevention & Control Policy